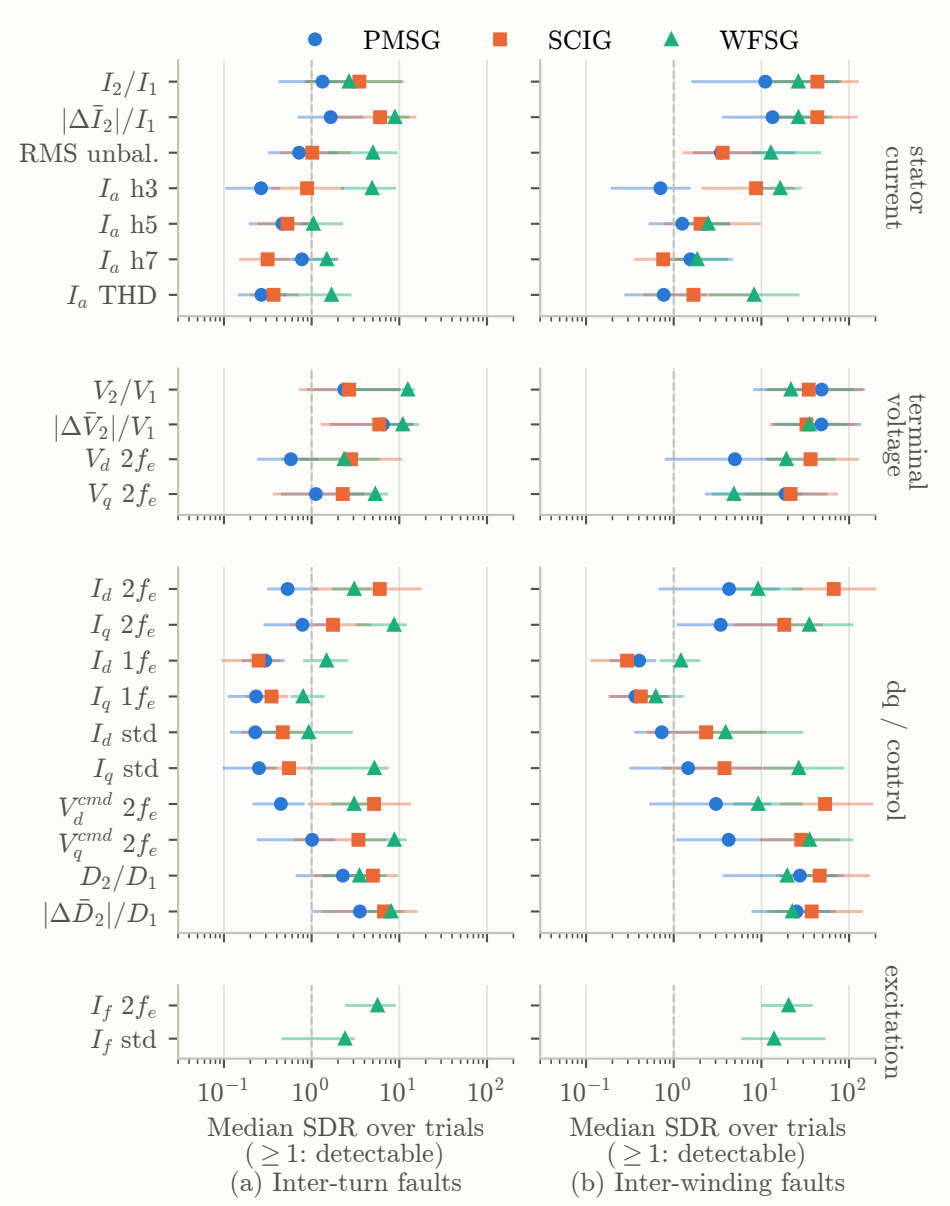
PMSG
SCIG
WFSG
I_2/I_1
$|\Delta\bar{I}_2|/I_1$
RMS unbal.
I_a h3
I_a h5
I_a h7
I_a THD
stator current
V_2/V_1
$|\Delta\bar{V}_2|/V_1$
V_d $2f_e$
V_q $2f_e$
terminal voltage
I_d $2f_e$
I_q $2f_e$
I_d $1f_e$
I_q $1f_e$
I_d std
I_q std
V_d^{cmd} $2f_e$
V_q^{cmd} $2f_e$
D_2/D_1
$|\Delta\bar{D}_2|/D_1$
dq / control
I_f $2f_e$
I_f std
excitation
10^{-1}
10^{0}
10^{1}
10^{2}
Median SDR over trials
(≥ 1: detectable)
(a) Inter-turn faults
Median SDR over trials
(≥ 1: detectable)
(b) Inter-winding faults